HangOut: Software Requirements Specification

Version 2.0 | Android (Kotlin + Firebase) | Replaces SRS v1.0

1. Introduction

1.1 Purpose

HangOut is a mobile app that helps people find nearby, like-minded people for spontaneous, casual hangouts such as sports, food, movies, walks, study sessions and trips. It is for friendship and shared activities, not dating. This document defines the complete requirements for the Android version, including everything built so far and every feature requested since v1.0.

1.2 Scope

- Platform: Android 7.0 (API 24) and above. iOS is out of scope for now.
- Users: anyone aged 15 or older.
- Core value: reduce boredom and isolation by connecting nearby people around a specific activity, right now or at a planned time.
- Launch strategy: one city or one campus first, to avoid the cold start problem.

1.3 Definitions

Term	Meaning
Hangout Card	A post describing an activity, time, place and group size that others can join
Join Request	A user tapping I'm in on a card
Hangout Chat	The group chat attached to one Hangout Card
Direct Message (DM)	A private one-to-one chat between two users, outside any card
Trust Score	A rating built from recent post-hangout reviews
Status	Done = built and tested, Partial = built but incomplete or buggy, Planned = not started, Paused = blocked by an external factor

2. Overall Description

2.1 Product Perspective

HangOut is a standalone Android app. The backend is Firebase: Authentication for sign-in, Cloud Firestore for data, and a file storage service for photos and videos. Google Play Services Location is used for GPS, and Google Maps for the map view.

2.2 User Classes

Class	Description
Regular User	Creates and joins hangouts, chats, comments, reviews, manages own profile
Hangout Host	A Regular User who posted a card; approves or declines join requests and can edit or cancel the card
Moderator / Admin	Reviews reports, removes content, bans accounts (initially the developer)

2.3 Operating Environment

- Android 7.0+ phones, tested on a Xiaomi device and the Android Studio emulator.
- Needs internet and GPS.
- Language: Kotlin. UI: XML layouts with Material Components.

2.4 Constraints and Assumptions

- Solo developer building over a summer, so features are delivered in phases (see section 10).
- Firebase free plan (Spark) is in use. SMS verification and possibly file storage can require the paid Blaze plan; see section 9.
- Users are assumed to have a data connection when using the app.
- Safety and moderation come before visual polish.

3. Current Build Status

Area	Status	Notes
Email and password sign-up and sign-in	Done	Account is created automatically if none exists
Email verification link	Done	Link often lands in spam during development
Phone OTP sign-in	Paused	Code is written; real SMS needs Blaze billing. Region policy already enabled
Google Sign-In	Planned	Button exists, shows a placeholder message
Auto-login on app start	Done	Checks session, email verified and profile complete
3-step profile setup	Done	First and last name, date of birth with age 15+ check, optional photo, interests
Profile saved to Firestore	Done	users collection
Feed with hangout cards	Done	List view; refreshes on resume
Map tab	Planned	Shows a coming soon message
Post hangout	Partial	Fixed chips only; no custom activity, date and time, custom size, description or images
Location detection	Done	GPS plus reverse geocoding to an area name
I'm in and join request saved	Done	Opens the hangout chat directly, without host approval
Hangout chat	Partial	Real-time text, emoji button and photo button; photos are not displayed as images; no delete
User profile view	Partial	Crashed on expired photo links; now guarded. Trust score is a placeholder
App theme	Partial	Light theme with coral and deep purple; some screens still inconsistent

4. Functional Requirements

4.1 Authentication and Identity

ID	Requirement	Priority	Status
FR-A1	User can sign in or sign up with one combined field that detects email or phone number	Must	Done
FR-A2	Email accounts must verify their email before entering the app	Must	Done
FR-A3	User can sign in with a phone number and a 6-digit OTP	Must	Paused
FR-A4	User can sign in with their Google account	Should	Planned
FR-A5	Accounts using different methods but the same person should be linkable, with phone number as the primary identity	Should	Planned
FR-A6	User can reset a forgotten password by email	Must	Planned
FR-A7	User can sign out, and can delete their account and all their data	Must	Planned
FR-A8	App reopens straight to the feed for a signed-in user with a complete profile	Must	Done
FR-A9	Optional selfie or photo verification badge	Could	Planned
FR-A10	Users under 15 are blocked at signup based on date of birth	Must	Done

4.2 Profile

ID	Requirement	Priority	Status
FR-P1	Profile has first name, last name, date of birth, age, optional photo and at least 2 interests	Must	Done
FR-P2	User can edit their profile any time: name, bio, photo, interests	Must	Planned
FR-P3	Profile has a short bio	Should	Planned

ID	Requirement	Priority	Status
FR-P4	Profile photo is uploaded to cloud storage and has a permanent link, not a temporary device link	Must	Planned
FR-P5	Any user can open another user's profile from a card, a comment or a chat	Must	Partial
FR-P6	Profile page shows photo, name, age, bio, interests, trust score, number of hangouts hosted and attended, and reviews received	Must	Partial
FR-P7	User can add photos to their own profile gallery, including photos from past hangouts or events	Should	Planned
FR-P8	User can control who sees their date of birth and last name; exact birth date is never shown to others	Must	Planned
FR-P9	Profile shows a Report and Block option on other users' profiles	Must	Planned

4.3 Posting a Hangout

ID	Requirement	Priority	Status
FR-H1	Host chooses an activity from preset tags	Must	Done
FR-H2	Host can type their own custom activity instead of using a preset	Must	Planned
FR-H3	Host picks an exact date and start time with a date and time picker, plus optional end time	Must	Planned
FR-H4	Host can also choose quick options: Right now, In 1 hour, This evening	Should	Done
FR-H5	Host can enter any group size, for example 2 to 50 people, instead of only two fixed options	Must	Planned
FR-H6	Host chooses the location by detecting GPS, searching an address, or dropping a pin on a map; the exact pin is only shown to approved participants	Must	Partial
FR-H7	Host can add a title and a description of up to 500 characters	Must	Planned

ID	Requirement	Priority	Status
FR-H8	Host can attach one or more photos to the card	Should	Planned
FR-H9	Host can set optional rules: same gender only, age range, verified users only	Could	Planned
FR-H10	Cards expire automatically after their end time and disappear from the feed	Must	Partial
FR-H11	Host can edit or cancel their own card; participants are notified of changes	Must	Planned
FR-H12	Host can mark a card as recurring, for example every Sunday morning	Could	Planned
FR-H13	Posting is rate limited, for example a maximum of 3 active cards per user	Must	Planned

4.4 Feed and Discovery

ID	Requirement	Priority	Status
FR-F1	Feed shows active cards as a list, newest or soonest first	Must	Done
FR-F2	Feed has a map view with a pin per card	Must	Planned
FR-F3	Cards show activity, host name, time, area name, spots left, and a thumbnail if there is a photo	Must	Partial
FR-F4	Feed can be filtered by activity type, distance, date and time	Must	Planned
FR-F5	Feed can be sorted by distance and by start time	Should	Planned
FR-F6	Search box finds cards by keyword in title, activity or description	Should	Planned
FR-F7	Pull to refresh and an empty state message when there are no cards	Must	Partial
FR-F8	Tapping a card opens a full detail page with description, photos, participants, comments and a join button	Must	Planned
FR-F9	Cards that are full, expired or cancelled are clearly marked or hidden	Must	Planned

4.5 Joining and Matching

ID	Requirement	Priority	Status
FR-J1	User taps I'm in to send a join request	Must	Done
FR-J2	Host sees pending requests with the requester's profile and trust score, and can approve or decline	Must	Planned
FR-J3	Only approved participants can enter the hangout chat and see the exact meeting point	Must	Planned
FR-J4	Spots left count updates when a request is approved; the card closes when full	Must	Planned
FR-J5	A user cannot join their own card or join the same card twice	Must	Partial
FR-J6	A participant can leave a hangout, and the host can remove a participant	Must	Planned
FR-J7	Chat suggests a public meeting spot, never a home address	Should	Planned

4.6 Chat and Direct Messages

ID	Requirement	Priority	Status
FR-C1	Each hangout has a real-time group chat for approved participants	Must	Done
FR-C2	Users can send private one-to-one direct messages from a profile, after a shared hangout or a mutual friend connection	Must	Planned
FR-C3	A messages inbox lists all hangout chats and DMs with the last message and unread count	Must	Planned
FR-C4	Messages can contain emojis, with a proper emoji picker	Should	Partial
FR-C5	Users can send photos that display as images inside the chat	Must	Partial
FR-C6	Users can send short videos that play inside the chat	Should	Planned

ID	Requirement	Priority	Status
FR-C7	Long press on a message offers Delete for me, which hides it only on this device account	Must	Planned
FR-C8	Long press on your own message offers Delete for everyone, which replaces it with a message deleted notice for all	Must	Planned
FR-C9	Messages show sender name and time; own messages are on the right, others on the left	Must	Done
FR-C10	Users can reply to a specific message and see delivered and seen status	Could	Planned
FR-C11	Typing indicator and online status	Could	Planned
FR-C12	Users can report a message and block the sender from the chat	Must	Planned
FR-C13	The message input bar always stays visible above the keyboard	Must	Done

4.7 Comments

ID	Requirement	Priority	Status
FR-M1	Any signed-in user can comment on a hangout card, visible to everyone who can see the card	Must	Planned
FR-M2	Comments show author name, photo, text and time; tapping the author opens their profile	Must	Planned
FR-M3	Users can delete their own comments; the host can delete any comment on their card	Must	Planned
FR-M4	Users can reply to comments and report abusive comments	Should	Planned
FR-M5	Host is notified when someone comments on their card	Should	Planned

4.8 Photos and Videos

ID	Requirement	Priority	Status
FR-X1	All photos and videos are uploaded to cloud storage and referenced by permanent links	Must	Planned
FR-X2	Images are compressed and resized before upload to save data and storage	Must	Planned
FR-X3	Uploads show progress and can be retried when they fail	Should	Planned
FR-X4	Participants can post photos from the hangout after it happens, shown on the card as an event gallery	Should	Planned
FR-X5	Video size and length are limited, for example 30 seconds and 25 MB	Should	Planned
FR-X6	Uploaded media can be reported, and moderators can remove it	Must	Planned

4.9 Reviews and Trust Score

ID	Requirement	Priority	Status
FR-R1	After a hangout ends, every participant is asked to review each other participant and the host	Must	Planned
FR-R2	A review has a thumbs up or down, 1 to 5 stars, quick tags such as fun, on time, felt safe, and an optional short written comment	Must	Planned
FR-R3	Only people who were approved participants of that hangout can review, and only once per person per hangout	Must	Planned
FR-R4	Trust Score is calculated from reviews, giving more weight to recent ones, and is shown on profiles and join requests	Must	Planned
FR-R5	New users show a New member label until they have enough reviews	Must	Done
FR-R6	Reviews on a profile can be read by other users; reviewers can be anonymous	Should	Planned

ID	Requirement	Priority	Status
FR-R7	Users can report an unfair or abusive review	Should	Planned
FR-R8	Review data also feeds matching, so people with similar ratings are shown to each other more	Could	Planned

4.10 Safety and Moderation

ID	Requirement	Priority	Status
FR-S1	Users can report any user, card, comment, message or photo with a reason category	Must	Planned
FR-S2	Reports are stored in a moderation queue; moderators can warn, remove content or ban a user	Must	Planned
FR-S3	Users can block others; blocked users cannot see their cards, message them or comment	Must	Planned
FR-S4	During a hangout, a user can share live location with a trusted contact	Should	Planned
FR-S5	In-app check-in and panic buttons during an active hangout	Should	Planned
FR-S6	Report categories include misuse of a friendship app for dating or unwanted romantic advances	Must	Planned
FR-S7	Optional same-gender-only filter on cards	Should	Planned
FR-S8	Firestore security rules restrict each action to the right user; for example only a host can edit their card and only the author can delete their message	Must	Planned

4.11 Notifications

ID	Requirement	Priority	Status
FR-N1	Push notification to the host for a new join request or comment	Should	Planned
FR-N2	Push notification to a user when their request is approved or declined	Should	Planned

ID	Requirement	Priority	Status
FR-N3	Push notification for new chat messages	Should	Planned
FR-N4	Push notification when a card matching the user's interests is posted nearby	Could	Planned
FR-N5	Reminder before a joined hangout starts and a review reminder after it ends	Could	Planned
FR-N6	Users can control notification types in settings	Should	Planned

4.12 Friends and Local Circle

ID	Requirement	Priority	Status
FR-L1	After a hangout, users can add each other as friends	Should	Planned
FR-L2	Friends can see each other's cards first and message each other directly	Should	Planned
FR-L3	Recurring hangouts and group communities for regular activities	Could	Planned

4.13 Settings

ID	Requirement	Priority	Status
FR-T1	Settings screen with edit profile, notifications, privacy, blocked users, sign out and delete account	Must	Planned
FR-T2	Terms of use and community guidelines are shown at signup and linked in settings	Must	Planned

5. Non-Functional Requirements

ID	Requirement
NFR-1	The feed loads within 3 seconds on a mid-range phone and a normal mobile connection

ID	Requirement
NFR-2	The app works on low-end Android 7.0 devices and on slow networks; images are loaded efficiently and cached
NFR-3	Exact user location is never shown to other users; only an area name, and the exact pin to approved participants
NFR-4	All data access is protected by Firestore and storage security rules; no collection is writable by every signed-in user without checks
NFR-5	Passwords are handled only by Firebase Authentication and never stored by the app
NFR-6	The app never crashes on missing, expired or invalid data such as an expired photo link; failures show a friendly message
NFR-7	Chat messages appear on other devices within 2 seconds on a normal connection
NFR-8	Reporter identity is hidden from the person reported
NFR-9	Users can export or delete their data on request
NFR-10	The UI uses the single approved theme and has readable contrast on every screen

6. User Interface Requirements

6.1 Theme: Golden Hour

Item	Value
Brand coral	#FF7E5F
Deep purple (toolbars, headings)	#2D2640
Warm amber (highlights)	#FFB86B
Screen background	#F7F5F2
Card and surface	#FFFFFF

Item	Value
Gradient (login and setup screens)	coral to dusk purple
Shape language	Rounded corners; pill buttons; cards with 16dp corners

6.2 UI Rules

- Light theme only for now; dark mode is not supported.
- Text must always be dark on light surfaces and light on dark surfaces. No white text on white backgrounds.
- Toolbars must sit below the status bar without overlap.
- Primary buttons use one consistent soft color; avoid harsh, high-intensity colors.
- Every screen has a clear back action.

6.3 Main Screens

Login, Profile Setup (3 steps), Feed (list and map), Hangout Detail, Post Hangout, Hangout Chat, Messages Inbox, Direct Message, User Profile, Edit Profile, Review, Notifications, Settings, Report.

7. Data Model (Firestore)

Collection	Key fields
users/{uid}	firstName, lastName, dobDay, dobMonth, dobYear, age, bio, photoUrl, interests, profileComplete, trustScore, reviewCount, createdAt
hangouts/{id}	postedByUid, postedByName, title, activityType, customActivity, description, startAt, endAt, locationLabel, lat, lng, maxPeople, currentPeople, photoUrls, rules, status, createdAt, expiresAt
hangouts/{id}/joinRequests/{uid}	userId, status (pending, approved, declined), joinedAt
hangouts/{id}/messages/{id}	senderId, senderName, type (text, image, video), text, mediaUrl, sentAt, deletedForEveryone, deletedFor (list of user ids)
hangouts/{id}/comments/{id}	authorId, authorName, text, parentId, createdAt

Collection	Key fields
hangouts/{id}/photos/{id}	uploaderId, url, createdAt
conversations/{id}	participantIds, lastMessage, lastMessageAt
conversations/{id}/messages/{id}	same fields as hangout messages
reviews/{id}	hangoutId, reviewerId, revieweeId, rating, thumbsUp, tags, comment, createdAt
reports/{id}	reporterId, targetType, targetId, reason, status, createdAt
blocks/{uid}/blocked/{uid}	blockedAt

8. Technology

Layer	Choice
Language and IDE	Kotlin in Android Studio
UI	XML layouts, Material Components, RecyclerView
Auth	Firebase Authentication (email and password, phone, Google)
Database	Cloud Firestore with real-time listeners
Media	Cloud file storage with image compression before upload
Location	Google Play Services Location and Geocoder; Google Maps SDK for the map view
Push	Firebase Cloud Messaging
Moderation backend	Firestore reports collection plus an admin tool, later Cloud Functions

9. Risks and Open Issues

Risk or issue	Impact	Plan
Real SMS OTP needs the paid Blaze plan	Phone login cannot be fully tested for free	Keep email login as the main method; use Firebase test phone numbers to finish and test the code; upgrade with a budget alert before launch
Photo upload and storage may also require Blaze billing for new projects	Profile photos, chat media and event galleries depend on it	Check the Firebase console before building FR-X1; if billing is not wanted, use a free third-party image host or limit media in the first release
Profile and chat photos currently use temporary device links	Photos break for other users and after restart	Build permanent uploads (FR-P4, FR-X1) before adding more media features
Firestore rules are too open	Any signed-in user can edit any hangout	Tighten rules (FR-S8) before any public testing
Verification emails land in spam	New users may think signup failed	Show a clear message; later use a custom email domain
Cold start: an empty feed has no value	Users leave	Launch in one campus or neighborhood and seed with real hangouts
One safety incident can end the app	Reputation	Reports, blocks, trust score and moderation are required before public launch
Scope is large for one developer	Project may not finish	Follow the phases below and ship a working core first

10. Delivery Roadmap

Phase	Goal	Contents
1: Stabilize (current)	Everything that exists works reliably	Fix theme on every screen, crash-proof profile view, show chat photos as images, tighten rules

Phase	Goal	Contents
2: Complete core posting	The post screen matches the real vision	Custom activity, date and time picker, any group size, location search and pin, title and description, card detail page
3: Real matching	Safe joining	Host approval, spots left, leave and remove, comments on cards
4: Media	Photos and videos that last	Cloud storage, profile photos, card photos, chat photos and videos, event galleries
5: Messaging	Full chat experience	Delete for me and for everyone, direct messages, inbox, emoji picker, unread counts
6: Trust and safety	Ready for strangers	Reviews, trust score, report, block, moderation queue, live location sharing
7: Reach	Keep people coming back	Push notifications, friends, filters and search, map view, Google sign-in, password reset
8: Launch	Real users	Settings, account deletion, terms, beta on one campus, Play Store release

11. Out of Scope for Version 1

- iOS app
- Payments, tickets or paid events
- Facebook login
- Public live streaming
- Voice and video calls
- Languages other than English
- Dark mode